

## Input / output

Reading inputs & outputs

Streams

→ Rivers

○ ○ ○ ○ ○  
Data infra → streams

System.out → object

PrintStream  
Class

print →

↓  
method

↓  
location

print  
Terminal

System.in

Scanner sc = new Scanner(System.in);

↓  
InputStream  
Class

nextInt()

→ get Integer

next() → String

nextLine() → entire line

nextBoolean()

hasNext() → Boolean  
method  
whether there is  
next input or not

It is present in java  
util package

Console → Read password

char[] ← read password  
→ security reason

Format Output

out.  
printf → Format String

%d → integer

%f → float

format method  
←

File Input & out

Path of ( "myfile.txt" ), Stream  
get path of file . . . . .

Read file

Big numbers

Precision

Integer

Numeric Types

'82' -> Long Beyond  
2 -> How to store

BigInteger

BigDecimal -> Store Floats &  
precision numbers

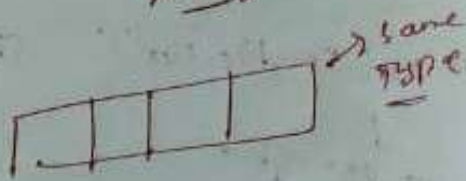
BigDecimal / BigInteger

valueOf() -> create  
number

multiply  
divide()  
add()  
subtract()

compareTo() -> compare

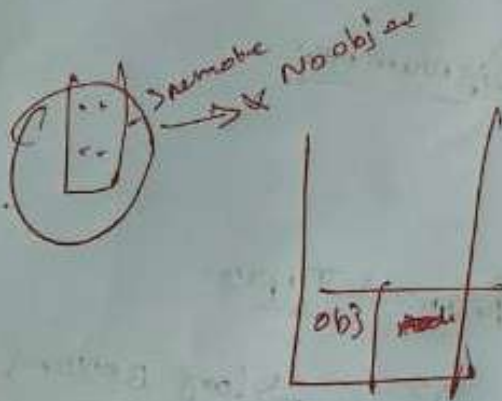
# Arrays



Declare array

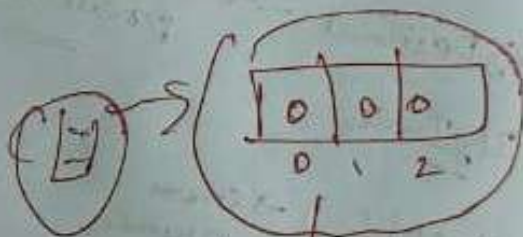
int arr;

Refer



int arr; → Declare array

new int[3];



index

Arr  
for



Access Array

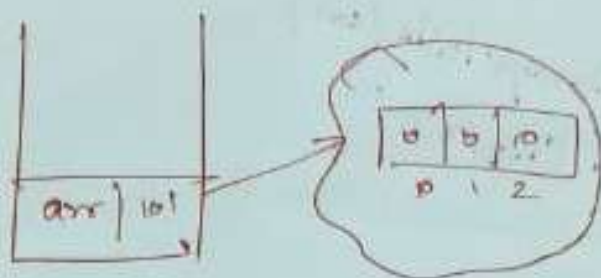


len

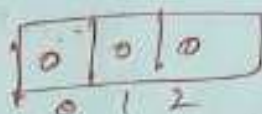
for-each

end





Access Array elem



indices

default value

object or string  
null → 0  
boolean → false

length = array length of arr

for - each loop

enhanced for - loop



int j

arr =

{ 1, 2, 3 }

with operator  
→ new



for (int ele : arr)

{

} → access them with index

tostrx()

Arrays tostrx(arr)

1 [1, 2, 3] ?

Array Copying

Print arr = {1, 2, 3};

strcpy brr = arr;

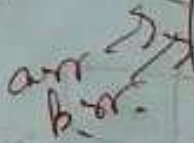


Both point to same memory

~~Array Copying~~

Char \* arr

brr[0] = 100;



arr[0]

↓

100

Arrays.copyOf()  
↓  
Copy arr element

Compile

javac Hello.java

Command line parameters

String args[]

run  
java Hello

argument passed

java Hello new word

Command line

Every Java class must have  
at least one class exactly one

public class same name as file name

Array Sort

a | b | c

1 | 2 | 3

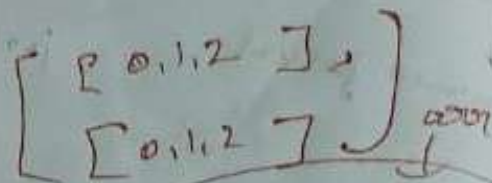
ascending order  
descending order } → sort  
Arrays.sort()



Multi-dimensional array



`int arr = new int[2][3];`



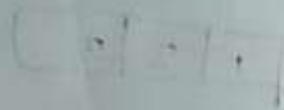
2 elements of 3rd  
array of arrays

`for (int i = 0; i < arr.length; i++)`

`{ for (int j = 0; j < arr[i].length; j++)`

`{`

→ each  
elem in row





## Ragged Arrays

int arr[3] = new int[3];

↓ null null null  
[ ~~0~~, ~~1~~, ~~2~~,  
null null  
~~0~~, ~~1~~ ]

]

[ null, null, null, null, null ]  
0 1 2 3 4

arr[0] = new int[1]; → {1}  
arr[1] = new int[2]; → {1, 2}

{ {1}, {1, 2}, {1, 2, 3} }  
{ {1, 2, 3, 4} }

Such arrays have diff column length  
are called Ragged Arrays

Java  
Array of array There is no  
such thing  
as multidimensional arrays

## Traditional programming

- \* Find actions (procedures) to do
- \* According to actions choose appropriate structures

doeg:- For eg. Making a tea  
Find methods of making tea / Store  
and then choose how to present a tea.

## Object-oriented

- \* Store the thing
- \* To do appropriate action on it

View it as Tea  
and then put ingredients by  
altering behaviour.

